From Raw Data to QGP Signatures: Real-Time Neural Network Event Selection



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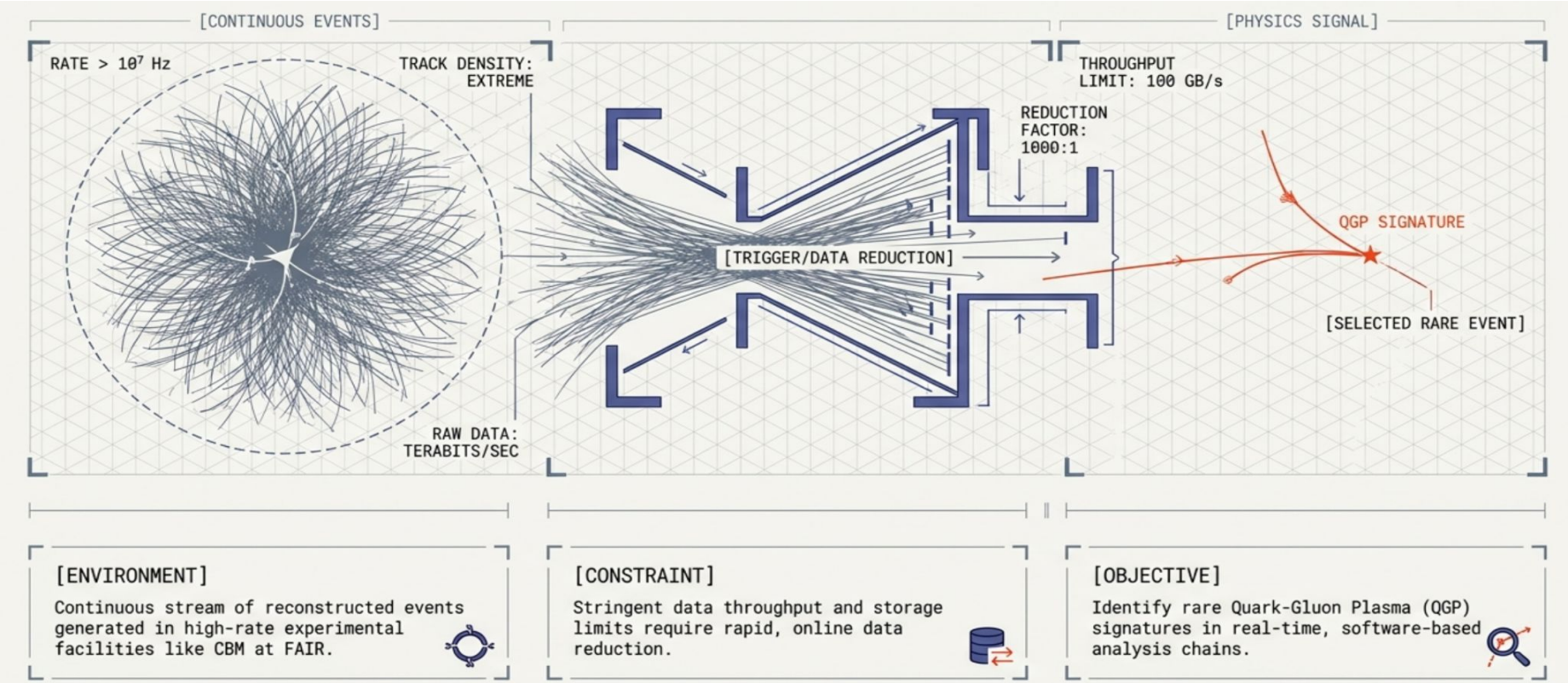
Based on 2605.26280
SHAP 2611.xxxx

QCD at extremes 2026
15.10.26

CRC-TR 211
Strong-interaction matter under extreme conditions

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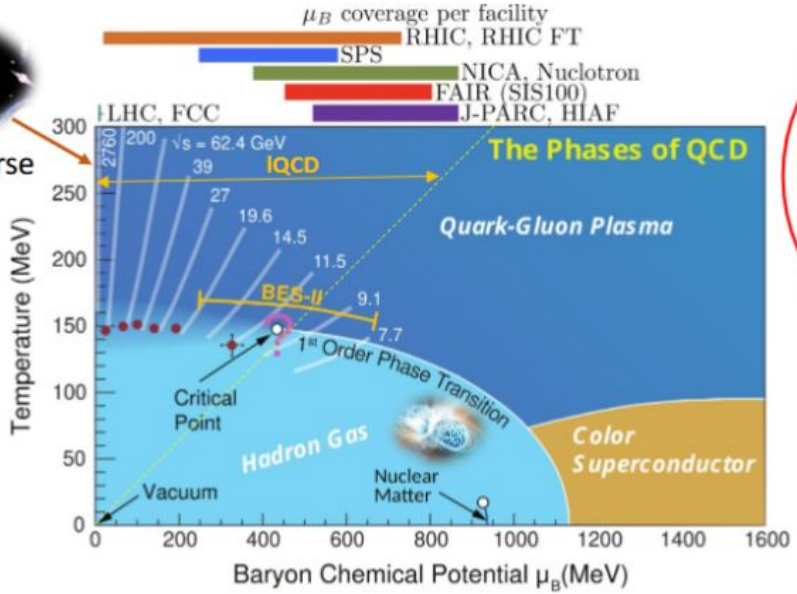
The high-rate trigger challenge

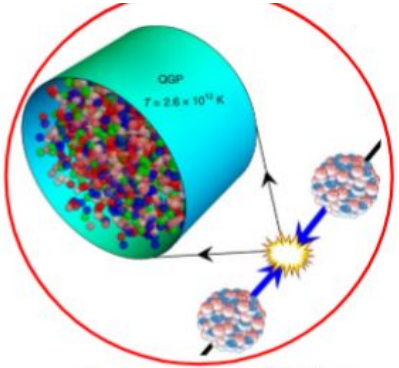


Quark-Gluon Plasma Search in HIC Experiments



Early Universe





heavy-ion collisions

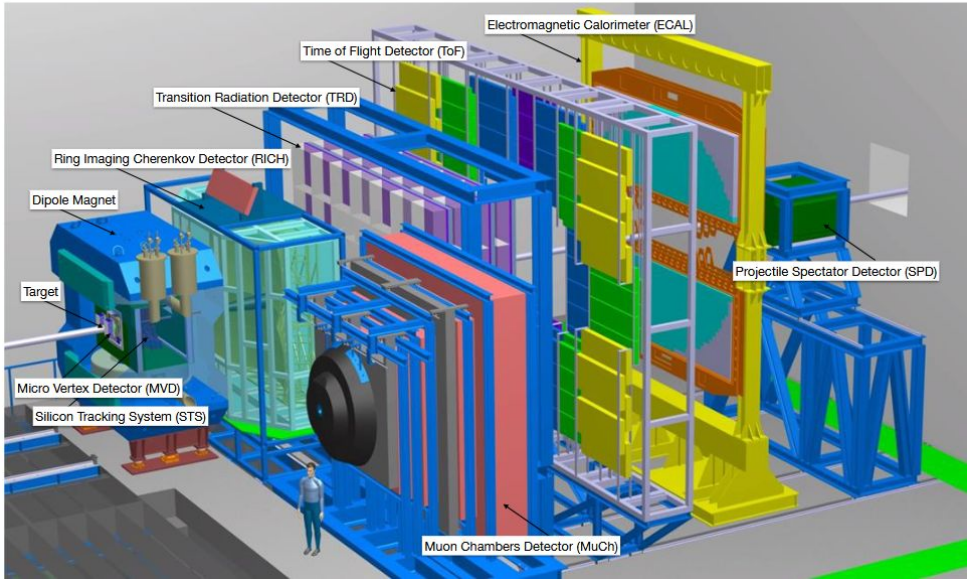
- Explore the QCD Phase Diagram
- Search for CEP and 1st order Phase Transition
- Quantify properties of created hot and dense QGP

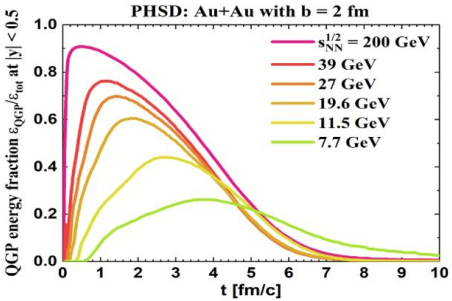
CBM experiment at FAIR

Main objective - to quantify the properties of created Quark-Gluon Plasma

! Due to the **extremely high rate,** only a small fraction of collisions can be stored for the analysis

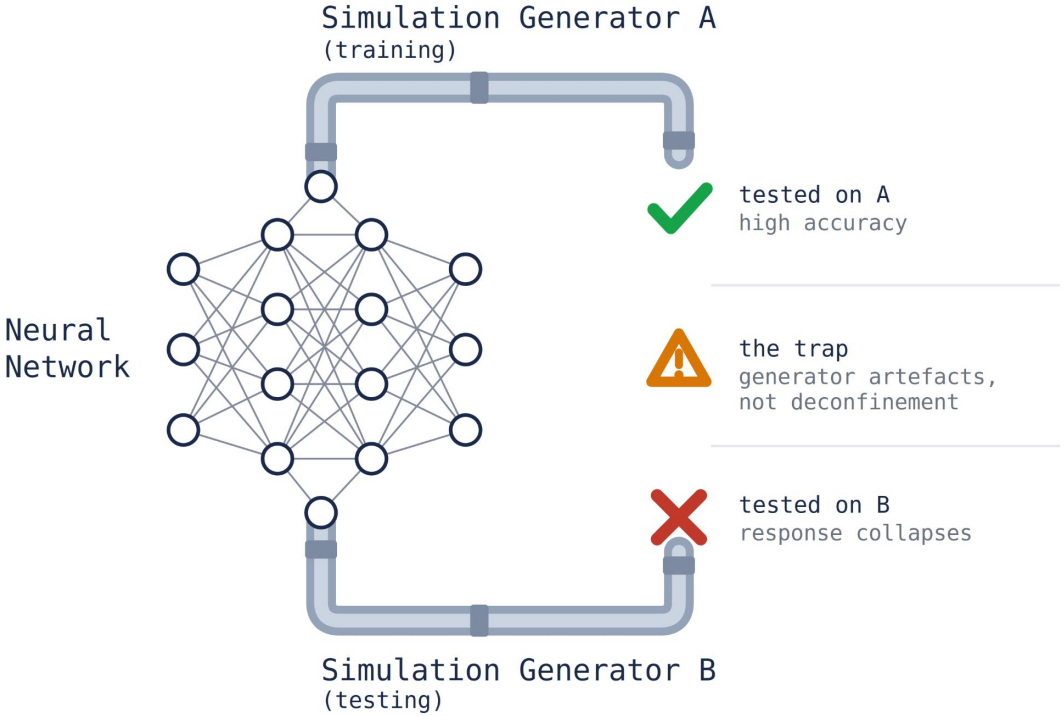
Real time data selection and triggering play a crucial role!





For CBM Physics - volume of QGP is essential
QGP fraction at FAIR energies are extremely small
Online QGP-sensitive selection is needed

Danger of Model Bias



THE TRAP

Neural Networks often learn model-specific correlations and simulation artifacts rather than true generic signatures of deconfined matter.

THE REQUIREMENT

Model independence is not an auxiliary property–it is a central requirement for experimental applicability.

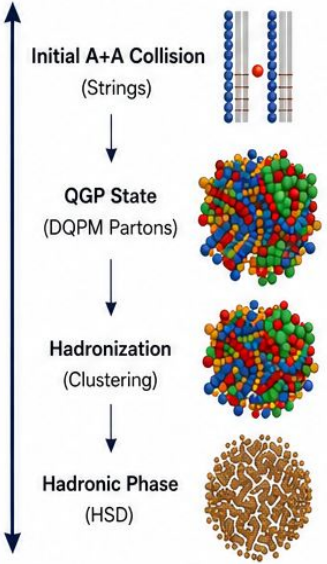
THE GOAL

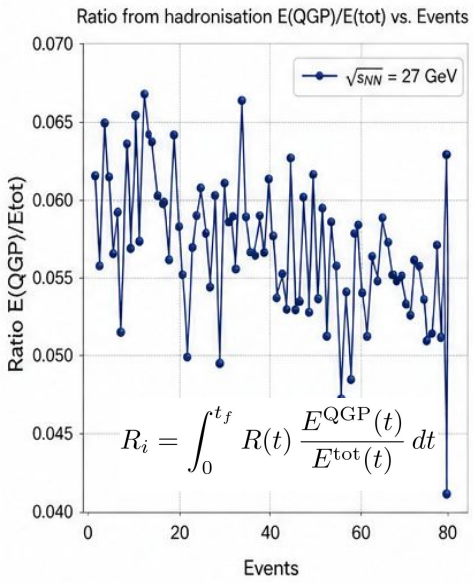
Build a classifier that extracts robust physical features, immune to the underlying physics assumptions of the training data.

Multi-model validation

PHSD (Parton-Hadron-String Dynamics)

Stages of Collision (PHSD)

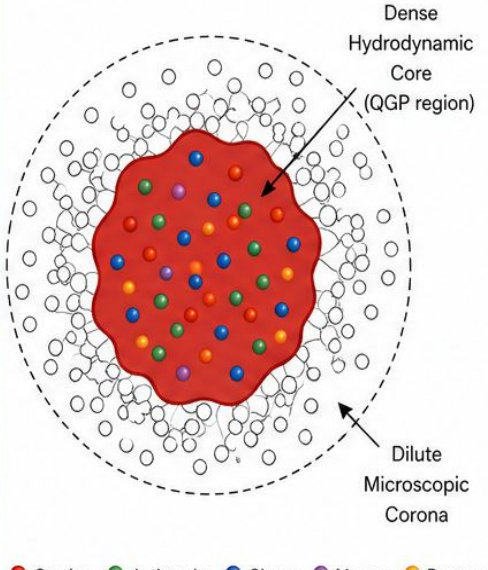




HSD - W. Cassing and E. L. Bratkovskaya, Nucl. Phys. A 831, 215 (2009); **PHSD 5** -P.Moreau, O.Soloveva,et al, PRC 2019

Microscopic off-shell transport providing **fully dynamical** event-by-event collision histories with **direct QGP labels** based on local energy density.

UrQMD (Ultra-relativistic Quantum Molecular Dynamics)



Key Idea

UrQMD models the dense, early stage of the collision with a hydrodynamic core, while the dilute outer region (corona) is treated microscopically.

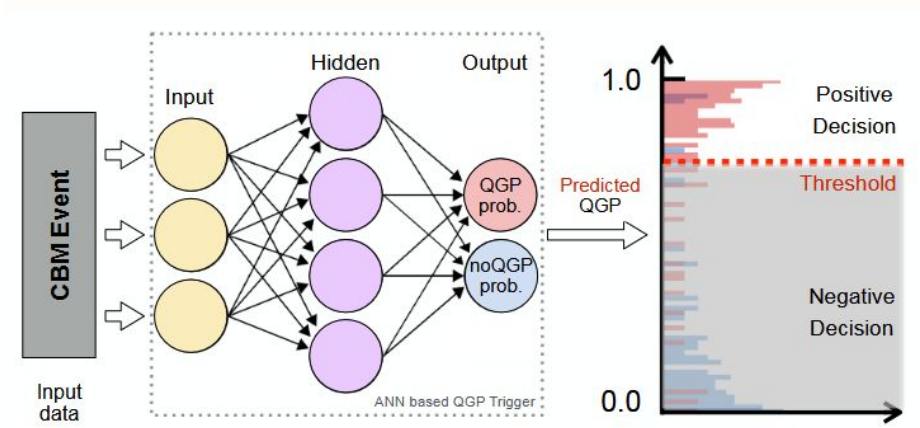
It provides **explicit QGP particle labels (quarks and gluons)** in the hydrodynamic core, enabling partonic degrees of freedom in the early stage.

Hadronic/hybrid transport. The QGP phase is modeled macroscopically in the **hydrodynamic core** (providing **explicit QGP particles: quarks and gluons**), while the dilute **corona** is treated microscopically.

UrQMD / core–corona J. Steinheimer and M. Bleicher, PRC **84**, 024905 (2011), arXiv:1104.3981.
UrQMD with chiral mean-field EoS:M. Omana Kuttan, A. Motornenko, et. al,Eur. Phys. J. C **82**, 427 (2022), arXiv:2201.01622

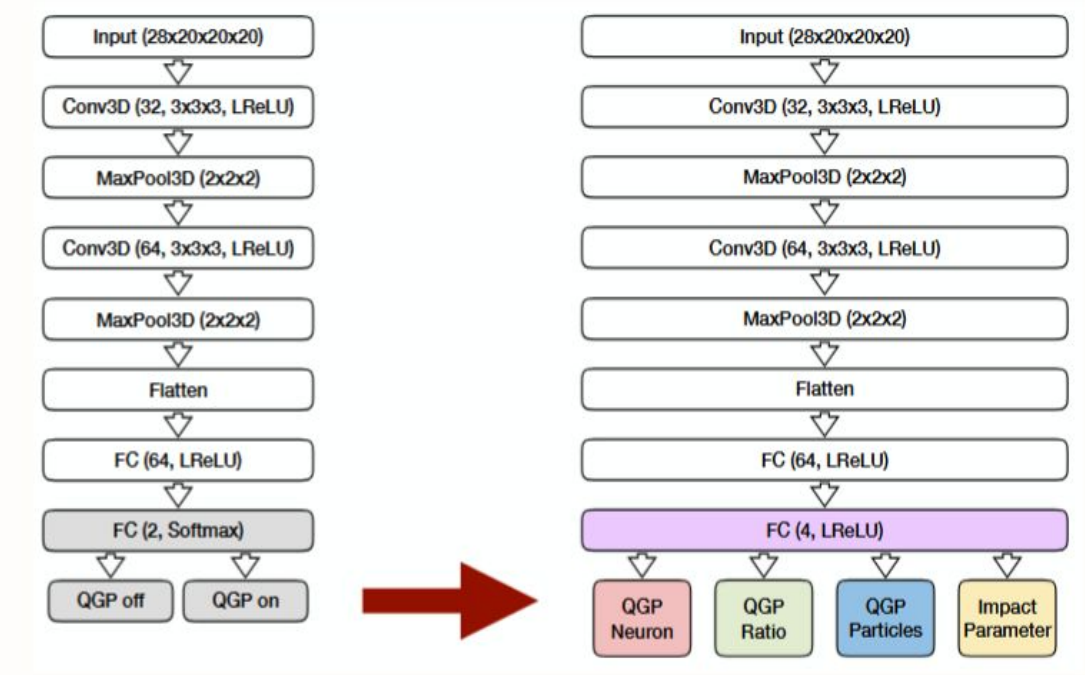
CNN- trigger concept for online QGP event selection in the CBM

- The NN processes reconstructed event data and predicts the probability that an event contains QGP
- A predefined threshold determines - the event is classified as QGP/noQGP
- Only events above the threshold are selected for storage and analysis
- Varying the threshold allows control of trigger sensitivity and selection efficiency



CNN scheme for PHSD

- The CNN model was updated to work with new PHSD data and predict several physical parameters for each event
- The output layer now has 4 neurons that give the QGP score, ratio, number of QGP particles, and the impact parameter.
- The activation function-**LeakyReLU**, which keeps the predicted values positive and physically realistic.
- The loss function was switched to MSE, making the training more stable for continuous outputs



Updated CNN architecture

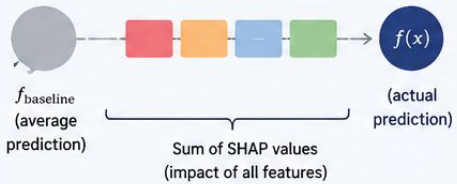
O.S., A.Belousov, I.Kisel. E. Bratkovskaya 2605.26280

SHAP (SHapley Additive exPlanations)

A game-theoretic approach to explain machine learning predictions

1. Intuition

SHAP values explain a prediction by measuring how much each feature contributes to moving the prediction from the average (baseline) output to the actual prediction.



2. The Idea (Shapley Value)

Imagine each feature as a player in a coalition game. We want to fairly distribute the "payout" (the prediction) among all players based on their contribution.



Payout = $f(x) - f_{\text{baseline}}$

(Total contribution above baseline)

3. How SHAP Values are Calculated

For each feature i, SHAP value ϕ_i is the average marginal contribution of that feature across all possible coalitions (all possible subsets of features).

Formula:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \left[f(S \cup \{i\}) - f(S) \right]$$

Where:

- N = set of all features
- S = a subset of features not containing i
- $f(S)$ = model prediction using features in S
- $f(S \cup \{i\}) - f(S)$ = marginal contribution of i to S
- The fraction is the weighting term from Shapley value theory

Example with 3 Features (A, B, C)

Coalition S (without i)		Add feature i		Marginal Contribution
{}		{i}		$f(\{i\}) - f(\{\})$
{A}	→	{A, i}	→	$f(\{A, i\}) - f(\{A\})$
{B}		{B, i}	→	$f(\{B, i\}) - f(\{B\})$
{A, B}		{A, B, i}	—	$f(\{A, B, i\}) - f(\{A, B\})$

Average (weighted) of above = ϕ_i

4. Final Explanation (Additive)

The prediction is explained as the sum of the baseline value and all SHAP values.

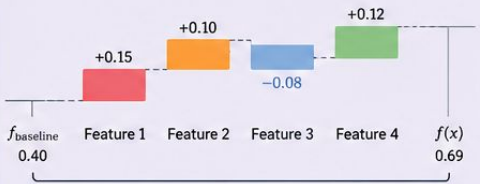
$$f(x) = f_{\text{baseline}} + \sum_{i=1}^{M} \phi_i$$

- f_{baseline}: Average prediction (baseline)
- $\sum_{i=1}^{M} \phi_i$: Sum of feature contributions (SHAP values)
- $f(x)$: Actual prediction

5. Interpretation

- **Positive** SHAP value ($\phi_i > 0$): feature pushes the prediction higher.
- **Negative** SHAP value ($\phi_i < 0$): feature pushes the prediction lower.

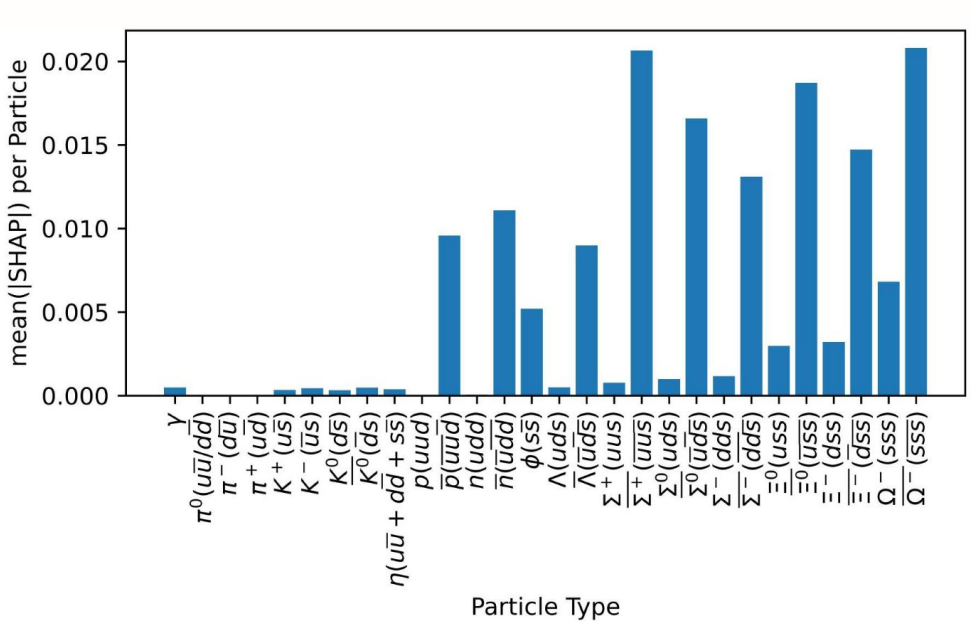
Visual Example (waterfall plot)



SHAP values sum to 0.29 (= 0.69 − 0.40)

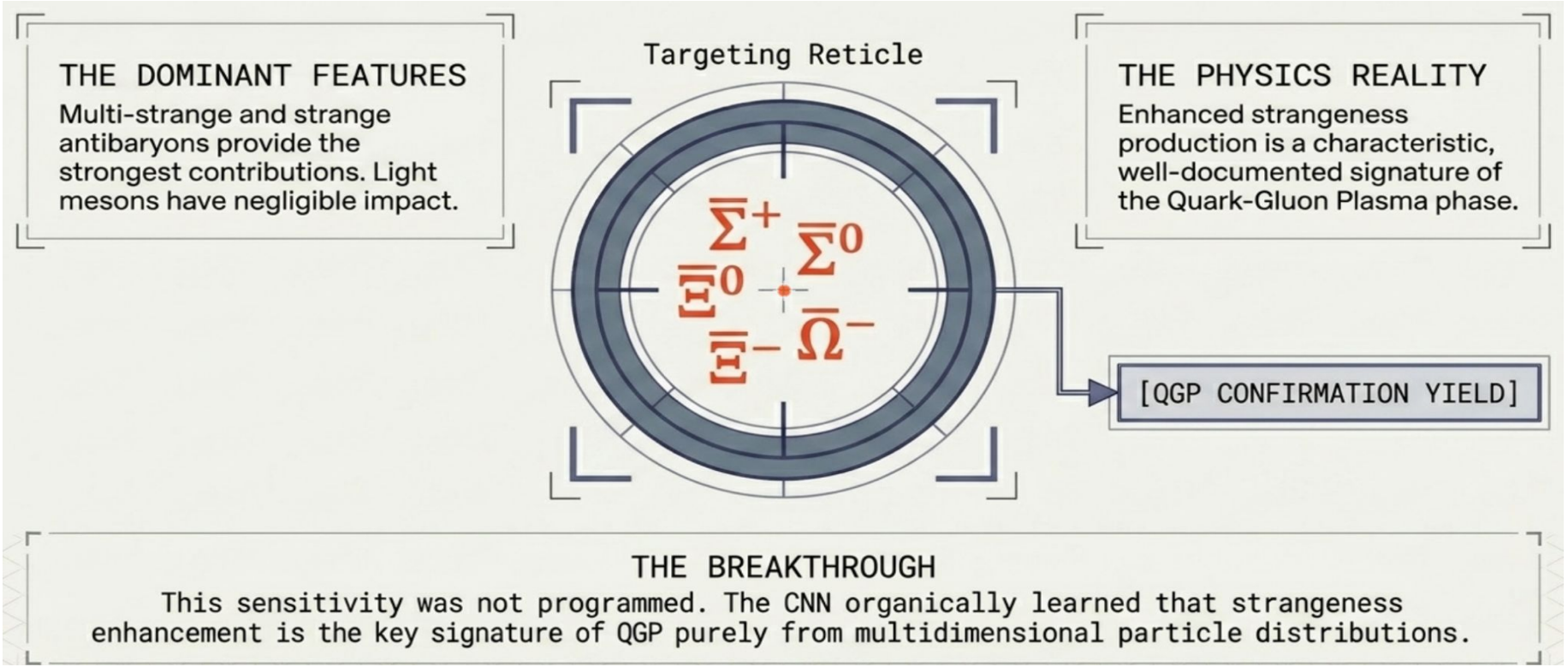
Shapley Additiv Explanations Values(SHAP)

- SHAP method to interpret the NN
- Each input feature acts as a player contribution to the model decision
- SHAP values shows how each feature influences the prediction toward QGP/non QGP classification
- This approach links ML decisions with physical QGP observables

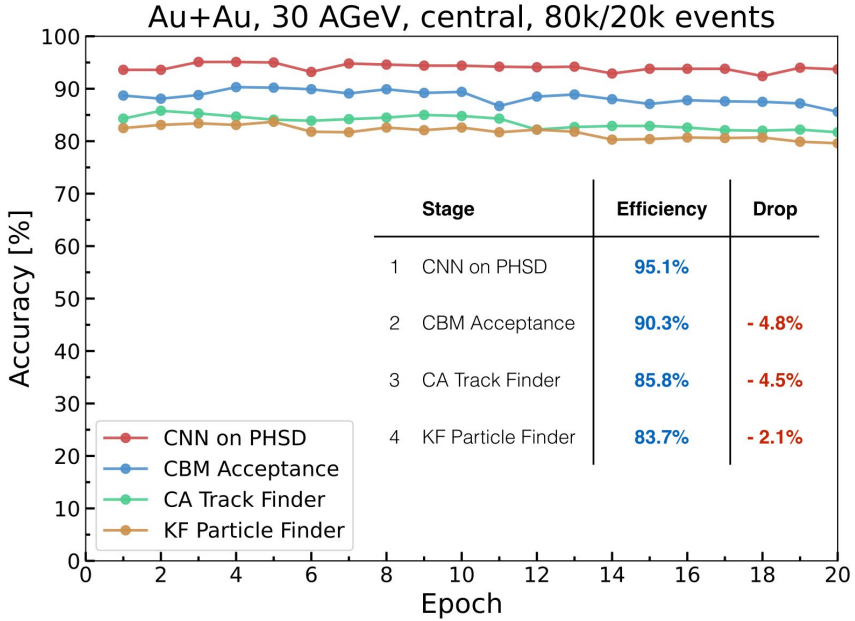


SHAP analysis highlights that strange and antybaryon observables contribute the most to QGP prediction

The Physics Fingerprint: Learning Strangeness

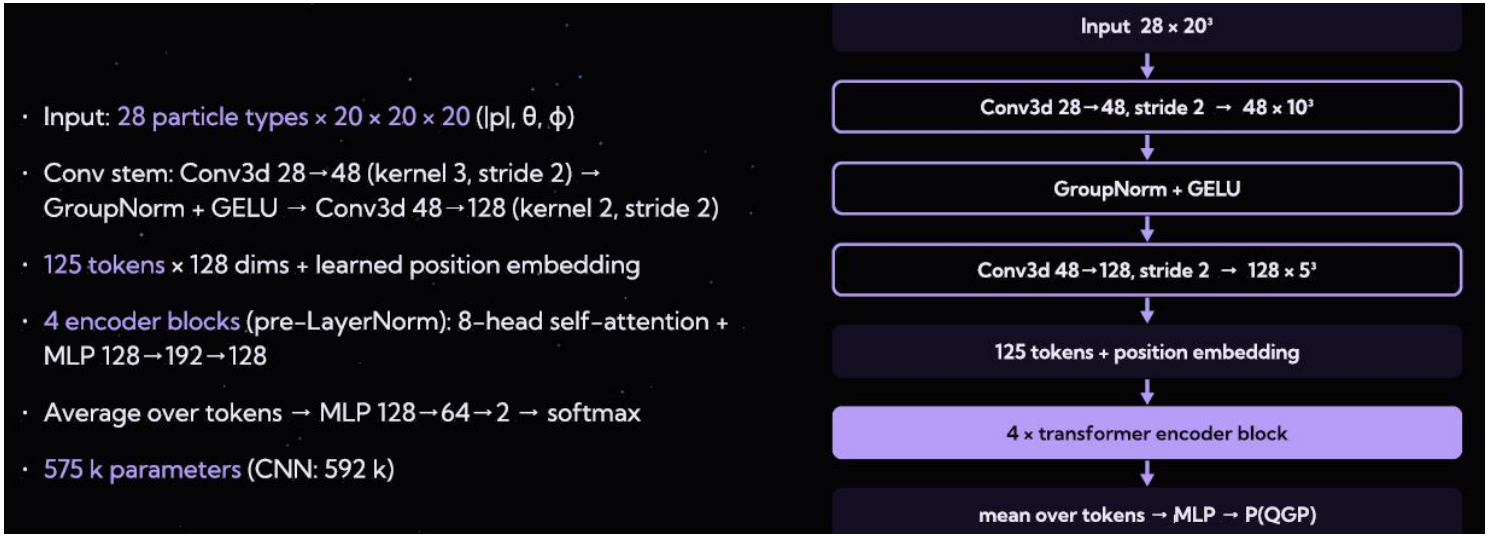


Classification accuracy in the CBM Experiment



- CNN keeps high resolution even after full reconstruction -> proves robustnes to detector effects and suitability for real time QGP event selection
- improving the network alone is insufficient unless the improvements survive reconstruction

Beyond CNN: Learning Global Correlations



- **Higher Accuracy:** Hybrid CNN–Transformer reaches **96.5%** accuracy (**25% fewer mistakes** than the CNN) on PHSD.
- **Model-Independent:** Retains **95.0%** accuracy on unseen PHQMD transport data, proving true physics-feature learning.
- **Real-Time Speed:** Exploits **99.3% data sparsity** with custom GPU kernels to reach **0.14 ms/event** ($4.4\times$ faster than CNN).
- **Next Decisive Step:** Validate the hybrid network through the realistic CBM detector & reconstruction chain.

Conclusions and Outlook

- CNN based event selection can identify collisions with QGP from reconstructed final/state particles
- High accuracy is preserved after full reconstruction -> robustness to detector effects and real/time applicability
- PHSD-UrQMD cross/validation demonstrates model independence
- SHAP analysis connects the NN response to physically meaningful particle features

Work in progress

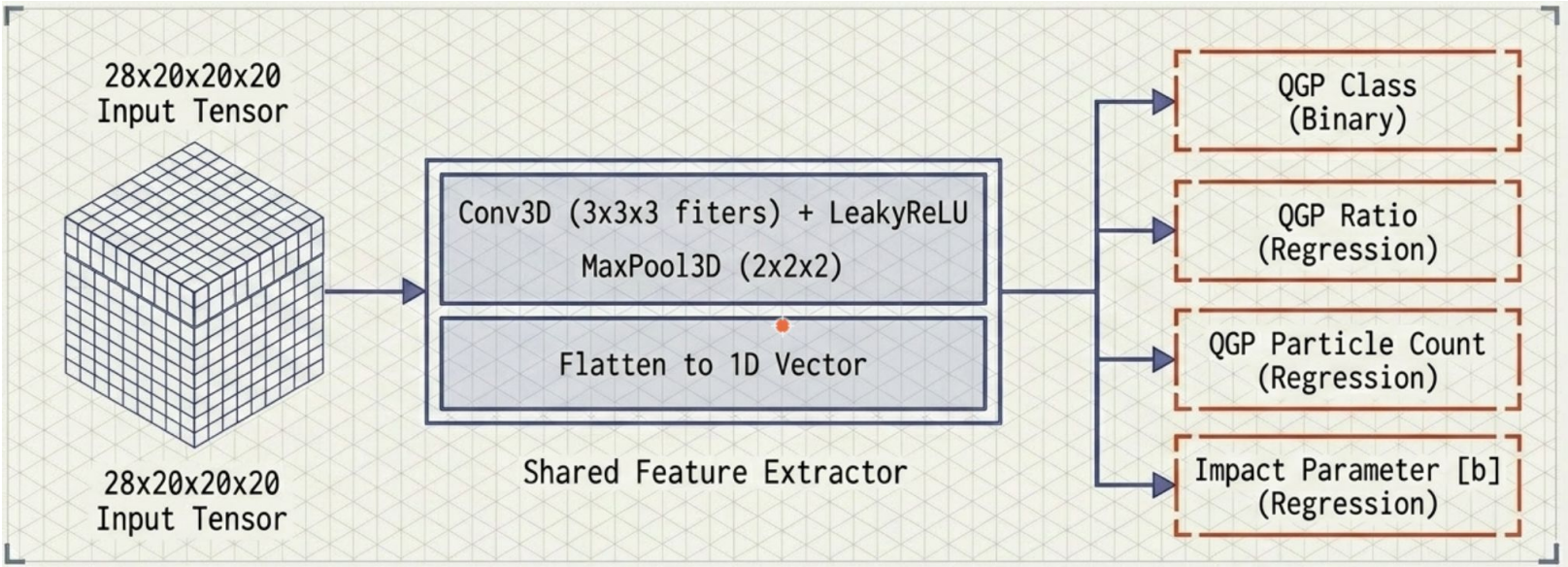
- testing on GPU
- GNN + CNN - similar results, now on Transformers
- Improvement from Physics/Simulations

Thank you for your attention!



olgasolovev.github.io

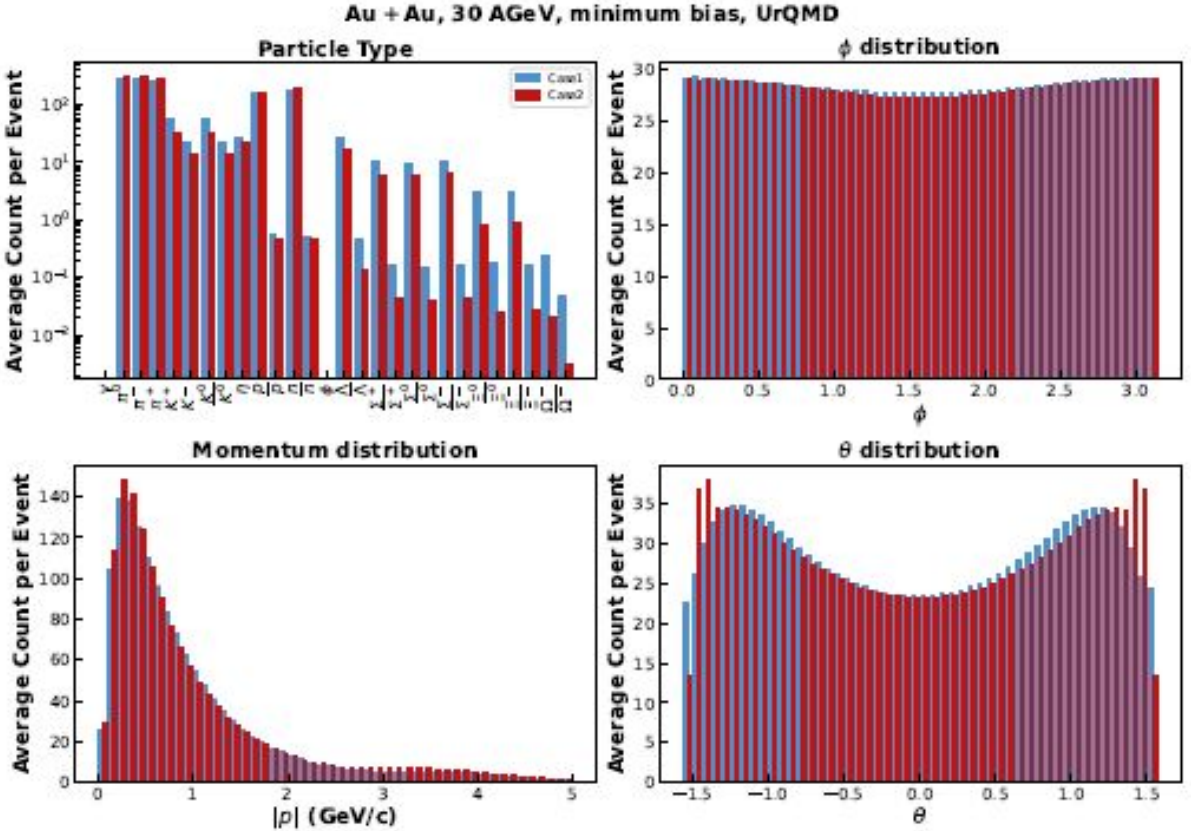
Input data



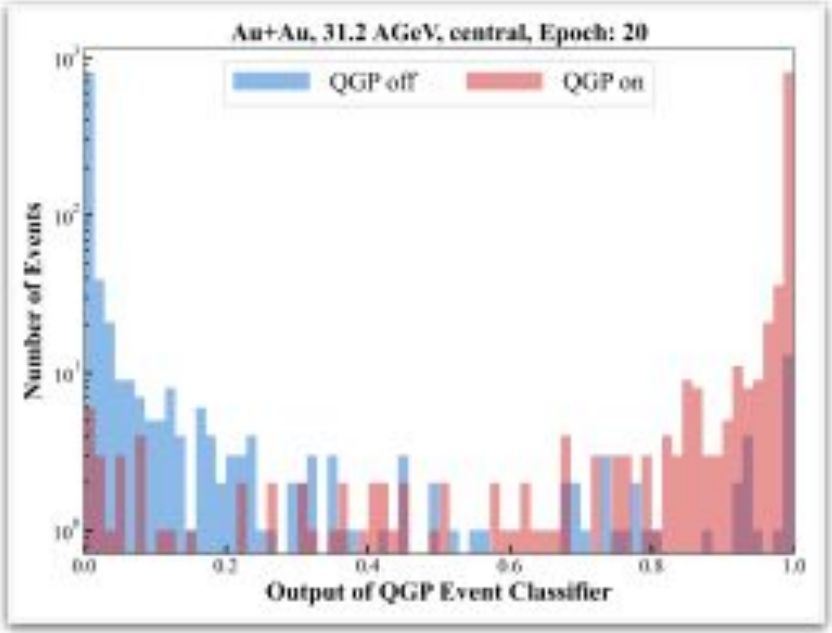
A single shared representation framework combining classification and regression directly from final-state data.

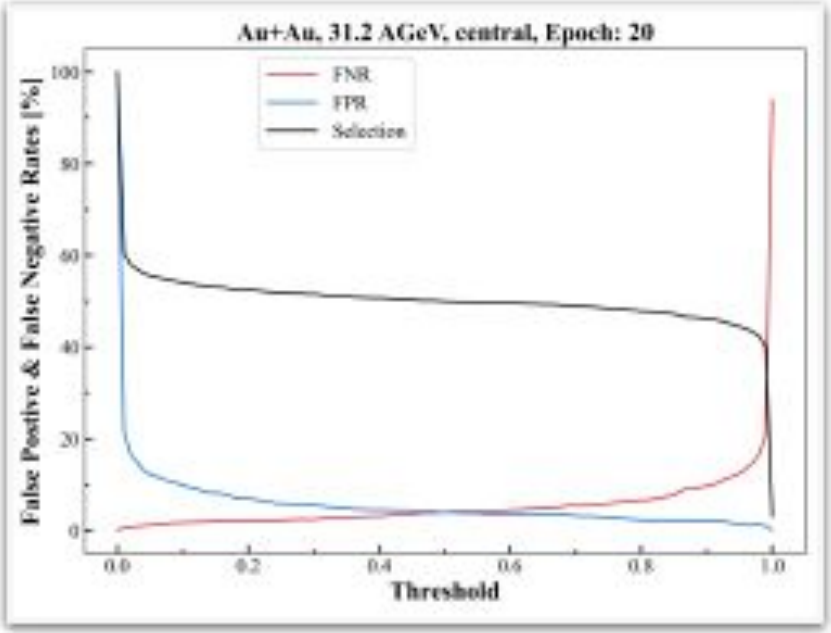
Average per-event input distributions

UrQMD:
Case 1 (hybrid evolution with hydrodynamic core, blue)
Case 2 (transport with chiral mean-field equation of state, red)

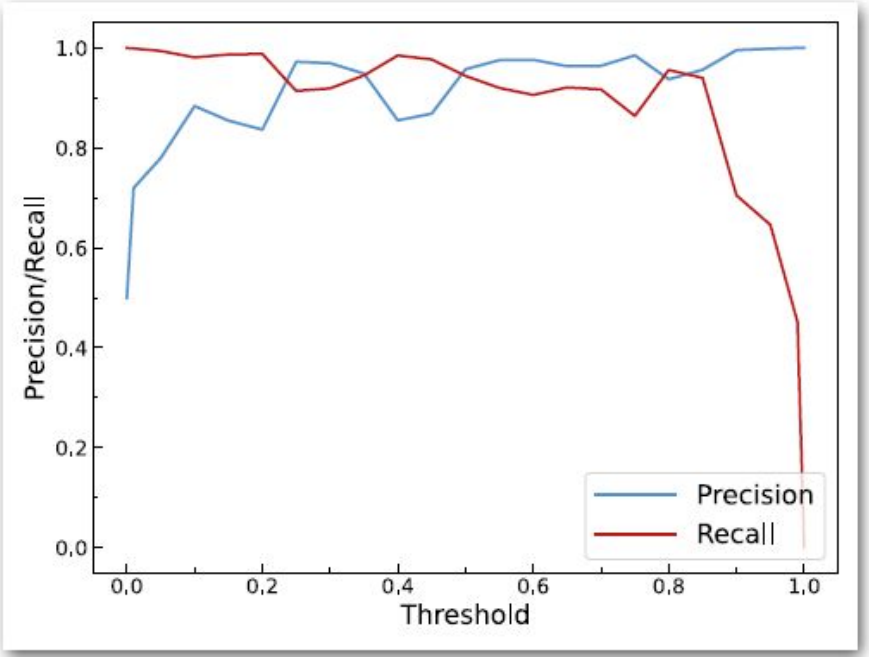


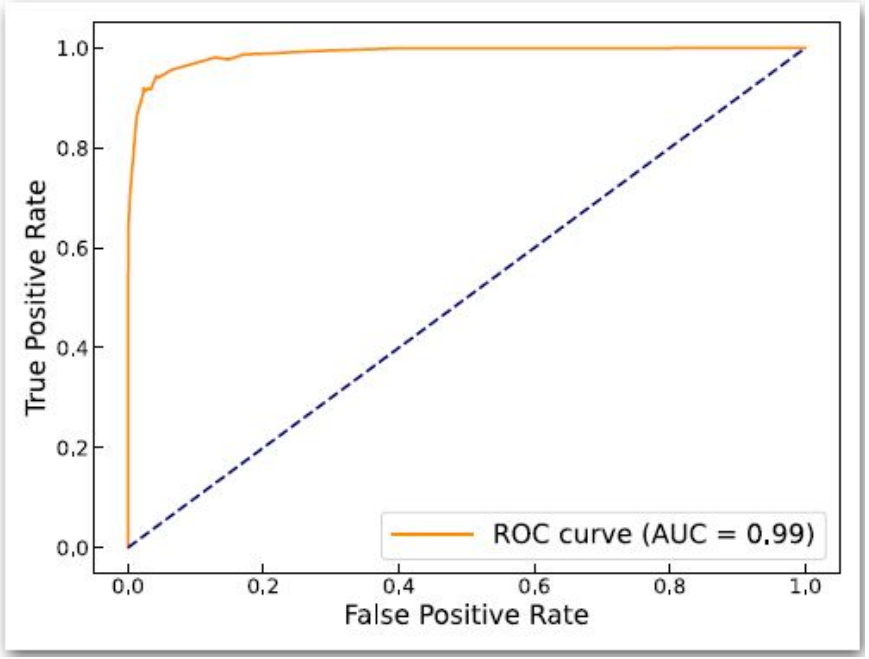
ROC


Au+Au, 31.2 AGeV, central, Epoch: 20
QGP off
QGP on
Number of Events
Output of QGP Event Classifier


Au+Au, 31.2 AGeV, central, Epoch: 20
FNR
FPR
Selection
False Positive & False Negative Rates [%]
Threshold

CNN - ROC


Precision/Recall
Threshold
Precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0
0,0
0,2
0,4
0,6
0,8
1,0


True Positive Rate
False Positive Rate
ROC curve (AUC = 0.99)
0.0
0.2
0.4
0.6
0.8
1.0
0,0
0,2
0,4
0,6
0,8
1,0

Multi-Model training and validation

